

Number System Interconversion

Number systems are the foundation of digital computing and mathematics. The four primary number systems we often use are:

- **Decimal (Base 10)**: Uses digits 0-9.
- **Binary (Base 2)**: Uses digits 0 and 1.
- **Octal (Base 8)**: Uses digits 0-7.
- **Hexadecimal (Base 16)**: Uses digits 0-9 and A-F.

Binary Conversions

1. Binary to Decimal

Steps:

1. Write the binary number.
2. Starting from the right, multiply each digit by 2 raised to its position index.
3. Sum the results.

Example: Convert 1101 to decimal:

$$\begin{aligned} & (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ &= 8 + 4 + 0 + 1 \\ &= 13 \end{aligned}$$

Decimal: 13

2. Binary to Octal

Steps:

1. Group the binary digits into sets of three starting from the right. Add leading zeros if necessary.
2. Convert each group of three into its octal equivalent.

Example: Convert 1101101 to octal:

```
110 1101 → 001 101 101 (grouped)
001 → 1
101 → 5
101 → 5
```

Octal: 155

3. Binary to Hexadecimal

Steps:

- 1. Group the binary digits into sets of four starting from the right. Add leading zeros if necessary.
- 2. Convert each group of four into its hexadecimal equivalent.

Example: Convert 1101101 to hexadecimal:

```
1101101 → 0001 1011 (grouped)
0001 → 1
1011 → B
```

Hexadecimal: 1B

Octal Conversions

4. Octal to Decimal

Steps:

- 1. Write the octal number.
- 2. Starting from the right, multiply each digit by 8 raised to its position index.
- 3. Sum the results.

Example: Convert 157 to decimal:

```
(1 × 8^2) + (5 × 8^1) + (7 × 8^0)
= 64 + 40 + 7
= 111
```

Decimal: 111

5. Octal to Binary

Steps:

1. Convert each octal digit into its three-bit binary equivalent.

Example: Convert 157 to binary:

```
1 → 001
5 → 101
7 → 111
```

Binary: 001101111

6. Octal to Hexadecimal

Steps:

1. Convert the octal number to binary.
2. Group the binary digits into sets of four starting from the right.
3. Convert each group into its hexadecimal equivalent.

Example: Convert 157 to hexadecimal:

```
Octal 157 → Binary 001101111 → Grouped 0001 1011
0001 → 1
1011 → B
```

Hexadecimal: 1B

Decimal Conversions

7. Decimal to Binary

Steps:

1. Divide the decimal number by 2.
2. Write down the remainder.

3. Repeat the division by 2 until the quotient is 0.
4. The binary number is the remainders read in reverse order.

Example: Convert 25 to binary:

```
25 ÷ 2 = 12, remainder = 1
12 ÷ 2 = 6,  remainder = 0
6  ÷ 2 = 3,  remainder = 0
3  ÷ 2 = 1,  remainder = 1
1  ÷ 2 = 0,  remainder = 1
```

Binary: 11001

8. Decimal to Octal

Steps:

1. Divide the decimal number by 8.
2. Write down the remainder.
3. Repeat the division by 8 until the quotient is 0.
4. The octal number is the remainders read in reverse order.

Example: Convert 100 to octal:

```
100 ÷ 8 = 12, remainder = 4
12  ÷ 8 = 1,  remainder = 4
1   ÷ 8 = 0,  remainder = 1
```

Octal: 144

9. Decimal to Hexadecimal

Steps:

1. Divide the decimal number by 16.
2. Write down the remainder (using A-F for values 10-15).
3. Repeat the division by 16 until the quotient is 0.
4. The hexadecimal number is the remainders read in reverse order.

Example: Convert 255 to hexadecimal:

```
255 ÷ 16 = 15, remainder = 15 (F)
15 ÷ 16 = 0, remainder = 15 (F)
```

Hexadecimal: FF

Hexadecimal Conversions

10. Hexadecimal to Decimal

Steps:

1. Write the hexadecimal number.
2. Starting from the right, multiply each digit by 16 raised to its position index.
3. Sum the results.

Example: Convert 1A3 to decimal:

```
(1 × 16^2) + (A × 16^1) + (3 × 16^0)
= 256 + 160 + 3
= 419
```

Decimal: 419

11. Hexadecimal to Binary

Steps:

1. Convert each hexadecimal digit into its four-bit binary equivalent.

Example: Convert 1A3 to binary:

```
1 → 0001
A → 1010
3 → 0011
```

Binary: 000110100011

12. Hexadecimal to Octal

Steps:

1. Convert the hexadecimal number to binary.
2. Group the binary digits into sets of three starting from the right.
3. Convert each group into its octal equivalent.

Example: Convert 1A3 to octal.

```
Hexadecimal 1A3 → Binary 000110100011 → Grouped 000 110 100 011
000 → 0
110 → 6
100 → 4
011 → 3
```

Octal: 0643